ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 2 · PRACTICE E · MARKING GUIDELINE

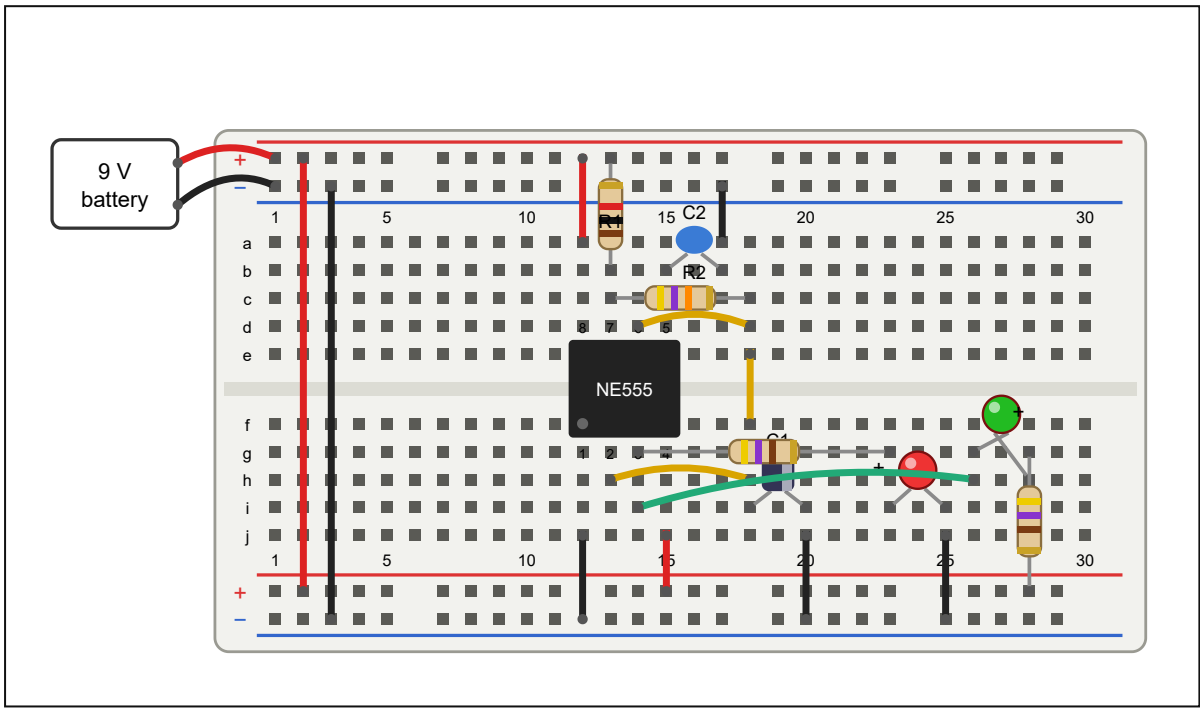
MARKS: 80

Marking guideline for the Academic Central NC(V) Level 3 Electronics and Digital Concepts for Robotics paper 2 · practice e.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Breadboard layouts may look different and still be correct: check that each connection in the circuit is made. Deduct 1 mark for each wrong connection and 2 marks for any short circuit (no less than 0 for a question).

QUESTION 1

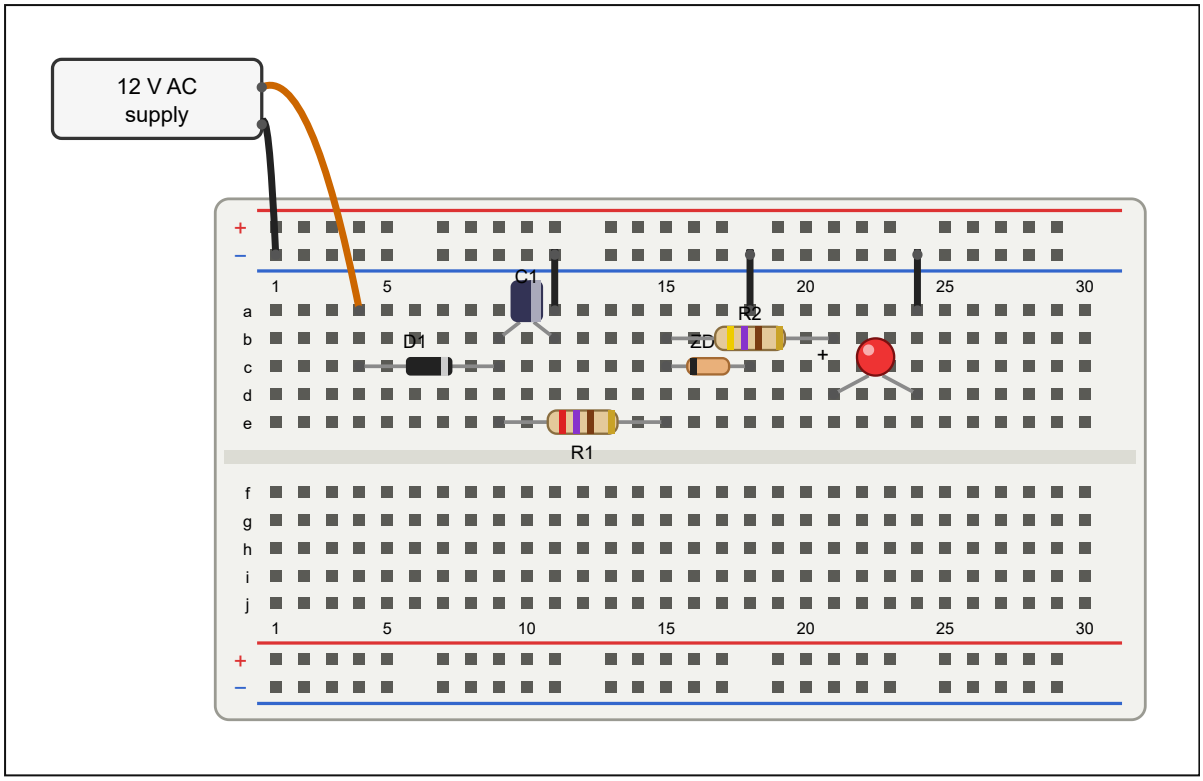


Example of a correct breadboard layout (other neat, correct layouts get full marks)

Placement: 555 across the centre gap ✓; R1 ✓; R2 ✓; R3 ✓; R4 ✓; C1 ✓; C2 ✓; LED1 ✓; LED2 ✓ = 9. Polarity: IC notch / pin 1 ✓; C1 ✓; both LEDs ✓ = 3. Connections ✓✓✓✓✓✓✓: pins 8 and 4 to +9 V; pin 1 to 0 V; R1 from +9 V to pin 7; R2 from pin 7 to pins 6 and 2 (joined); C1 from pins 6 / 2 to 0 V; C2 from pin 5 to 0 V; pin 3 through R3 and LED1 to 0 V; LED2 and R4 from +9 V to pin 3 = 8. (20)

[20]

QUESTION 2

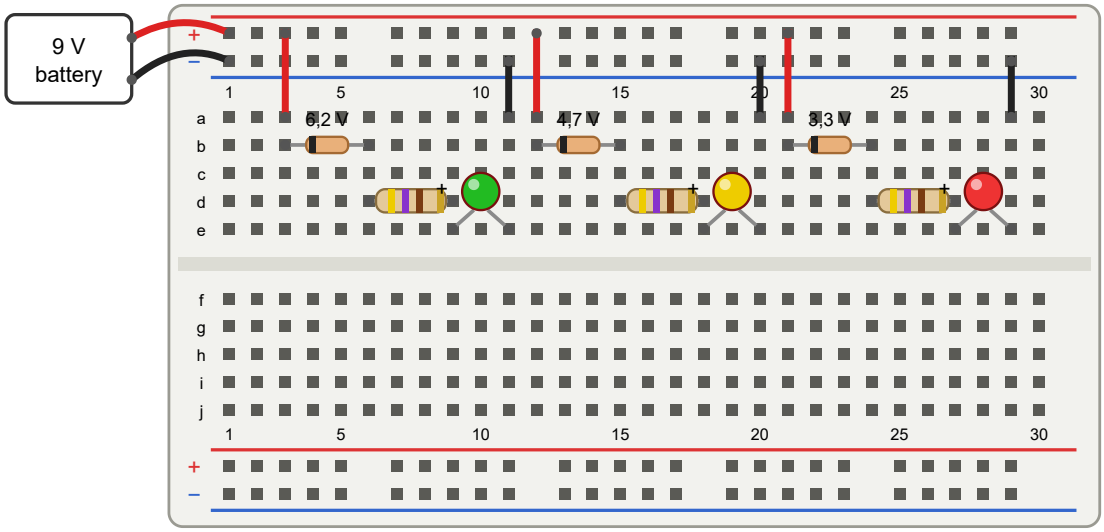


Example of a correct breadboard layout (other neat, correct layouts get full marks)

Placement ✓✓✓✓✓ (diode, C1, R1, Zener, R2, LED) = 6. Polarity ✓✓✓: diode (cathode towards C1); C1 + to the rectified side; Zener cathode to the positive line (reverse biased); LED = 4. Connections ✓✓✓✓✓✓✓✓✓: AC wire 1 to the diode anode; diode cathode to the + line; C1 across the + and 0 V lines; R1 from the + line to the regulated point; Zener from the regulated point to 0 V; R2 from the regulated point to the LED; LED cathode to 0 V; AC wire 2 to 0 V; supply wires labelled; neat, no short circuits = 10. (20)

[20]

QUESTION 3



Example of a correct breadboard layout (other neat, correct layouts get full marks)

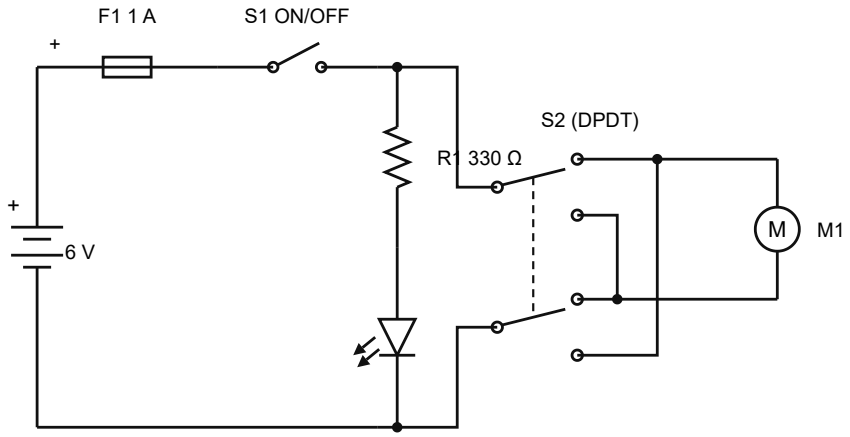
Correct circuit: three identical branches across the 9 V battery; in each, the Zener cathode goes to +9 V, its anode to a 470 Ω resistor, the resistor to the LED anode, and the LED cathode to 0 V (6,2 V Zener with green, 4,7 V with yellow, 3,3 V with red).

Marks, 2 each: battery with correct polarity ✓✓; three separate branches in parallel across the battery ✓✓; 6,2 V Zener with the green LED ✓✓; 4,7 V Zener with the yellow LED ✓✓; 3,3 V Zener with the red LED ✓✓; Zeners reverse biased (cathode to +) ✓✓; one 470 Ω resistor in series in each branch ✓✓; LEDs the right way round ✓✓; neat use of rails and jumpers ✓✓; printout with name / examination number ✓✓ (20)

[20]

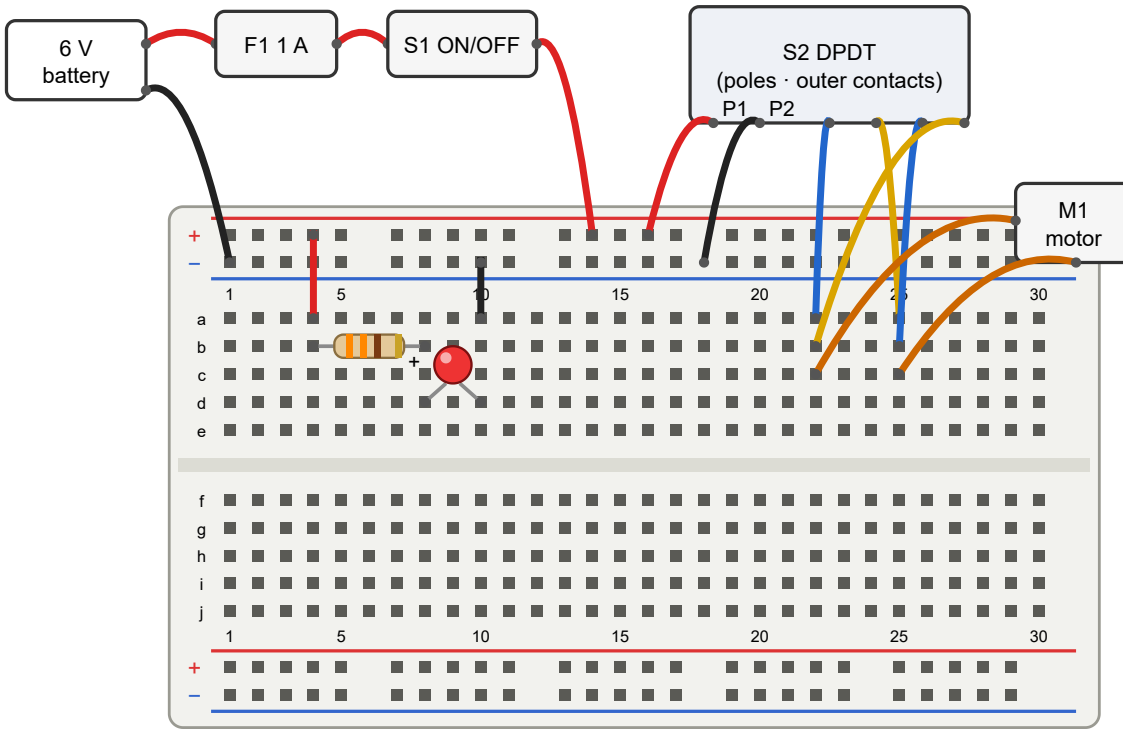
QUESTION 4

4.1



(Battery with polarity 1; fuse and ON/OFF switch in series with the supply 1; DPDT switch with the supply on the poles and cross-wiring between the outer contacts 1; motor across one pair of outer contacts 1; LED with 330 Ω across the supply after the ON/OFF switch, correct symbols 1)

(5)



Example of a correct breadboard layout for 4.2

4.2 Placement ✓✓✓✓✓✓ (battery, fuse, ON/OFF switch, DPDT switch, motor, LED, resistor) = 7. Polarity ✓✓: battery; LED = 2. Connections ✓✓✓✓✓: fuse in series from battery +; ON/OFF switch after the fuse; supply to the DPDT common pins; cross-wiring of the outer pins (reversing); motor to the DPDT outputs; LED and resistor across the switched supply = 6. (15)

[20]

TOTAL: 80

END OF MARKING GUIDELINE